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Fiscal transfers and inflation: Evidence from India

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Abstract

Controlling for monetary policy, government transfers are potentially inflationary. This, however, may not be true when the economy is demand-constrained. Using a panel data of 17 Indian states over 30 years, we show that government transfers via welfare programs do not lead to inflation. For identification, we use a narrative shock series of transfer spending that is based on the introduction of new welfare programs. We re-examine the relationship between government transfers and inflation by studying whether the recent implementation of India's public workfare program, NREGA, had aggregate price effects. Using the phase-wise implementation design of the program, we confirm the absence of any association between higher program coverage and price-inflation.

JEL: E31; E62; H53; I38

Keywords: Fiscal Transfers; Welfare programs; Government spending; Inflation

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1 Introduction

A wide variety of welfare measures primarily consist of fiscal transfers. Owing to the long lags associated with the design, approval, and implementation of large government purchase programs, redistributive fiscal transfers are often an attractive policy response to economic downturns, especially in developing economies. However, concerns regarding fiscal transfers fueling price inflation can often impede such policy interventions. In an economy that is near the efficiency frontier, large scale transfers can boost demand, thus accelerating inflation (Bilbiie et al., 2013). Within categories of government spending, government transfers are often viewed as more inflationary than government purchases. The argument goes that, given a fixed monetary policy stance, while both transfers and purchases boost aggregate demand, only government purchases increase aggregate supply (see for example Giambattista and Pennings, 2017).

Recent literature on government purchases multiplier has indeed established the expansionary nature of government purchases. As example, Acconcia et al. (2014) instrument temporary contractions in public spending at the province-level in Italy on evidence of mafia involvement to estimate the government spending multiplier between 1.5-1.9. Serrato and Wingender (2010) use the census shock due to measurement error in US census estimates as an instrument to estimate the government spending multiplier to be around 1.88. Fishback and Kachanovskaya (2010) study the multiplicative effects of government grants in US states during the New Deal using a swing voting measure as an instrument. They report an output multiplier of 1.67 corresponding to grants on public work and relief. Nakamura and Steinsson (2014) estimate an "open economy relative multiplier" of around 1.5.

Government transfers, on the other hand, are output neutral if we assume a representative agent model with lump sum taxes and transfers. In a recent study, however, Oh and Reis (2012) show how "well targeted" transfers can have expansionary effects on output. Transfers can also change marginal incentives by changing relative prices: subsidizing productive assets to rural families below the poverty line, for example, can increase the marginal incentive of transfer beneficiaries to acquire productive capital. An increase in marginal reward might spur economic activity.[a] Finally, Woodford (1990) shows how government transfers can lead to higher investment and output by alleviating liquidity constraints.

Government transfers, therefore, can also stimulate both demand and supply. This is especially true for developing economies where product markets are often demand constrained which can cause fiscal transfers to be expansionary without increasing price inflation. There is some recent evidence to suggest this. Taking evidence from a large food-assistance program in Mexico, Cunha et al. (2018) show that while in-kind transfers reduced prices, cash transfers had no effect on prices. Similarly, Egger et al. (2019) show that in Kenya, a large fiscal transfer ($\approx$ 15% of local GDP) had an output multiplier of 2.6, while causing an economically insignificant price inflation of 0.1%. While, these studies use specific programs as evidence, in this paper we study welfare spending in India over a period of 30 years to examine the effect of government transfers on inflation.

A key challenge in successfully identifying the effect of government transfers on inflation is that this relation is often confounded by contemporaneous changes in the macroeconomic environment and monetary policy. Ignoring these issues can lead to spurious correlations between government transfers and inflation. The problem is further compounded for developed economies where a substantial proportion of government transfers are in the form of automatic stabilizers. We circumvent these issues by using discretionary changes in transfer spending in India due to new program introductions. Further, since we use sub-national

[a]On the other hand, a decrease in marginal reward by the way of less return to working or saving may dampen output and employment.

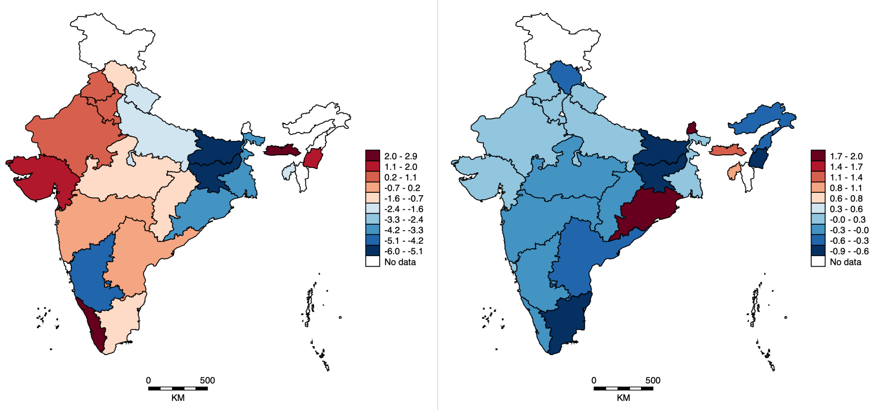

Figure 1: Variation in inflation (left) and changes in fiscal transfer (right) across states in year 2000

variation in transfer spending in India, this allows us to control for aggregate fiscal and monetary policy together with any aggregate macroeconomic shocks. India is particularly well suited for an analysis at the sub-national level because of (i) the federal structure of the government with substantial within and across states variation in transfer spending and (ii) the large geographic size of the country with substantial differences in prices across states.

An analysis of variance for the annual rate of inflation at the state-level (controlling for year fixed effects) shows that while changes in the country-wide average inflation explain about 60% of the variation, the remaining 40% is attributable to variation between states. The left panel of Figure 1 shows the variation for year 2000 as an example. Similarly, if we look at year-on-year changes in fiscal transfer for each state normalised by the agricultural output of the state, country-wide changes explain less than 25% of the variation in the series. The right panel of Figure 1 illustrates the inter-state variation. Hence, due to the reasons mentioned above there is enough variation in both the regressor and the regressand even at a sub-national level.

Another key challenge is that changes in transfer spending may be endogenous to other macroeconomic changes, such as economic downturns and recessions, which can also directly affect inflation. Ignoring this issue can spuriously over/under-report the effect of government transfers on prices. In general, the literature follows three key identification strategies to circumvent the problem of endogeneity. First, is the VAR-based approach. See for example, Blanchard and Perotti, 2002; Mountford and Uhlig, 2009; Ilzetzki, Mendoza, and Végh, 2013 to see how different VAR-based approaches achieve identification using different structural assumptions regarding the dynamics of inflation, output, and fiscal policy. The second approach is to find an external source of variation in fiscal policy – an instrument – that is uncorrelated with the rate of inflation or with factors that may affect the rate of inflation. The third strategy involves isolating a subcomponent of spending that is independent of the current macroeconomic environment and aggregate price dynamics. Barro (1981), for example, argued that military expenditure in the US during major wars can be regarded as exogenous as these conflicts were not correlated with aggregate business cycles in the US. Similarly, Ramey and Shapiro (1998) and Ramey (2011) use news reports to identify changes in government spending due to military build-ups that are independent of the state of the economy. Romer and Romer (2010, 2016) use narrative analysis to identify exogenous changes in taxes and transfers to estimate their impact on economic activity in the US.

Following Bahal (2019), this article uses the narrative analysis approach of identifying exogenous changes in transfer spending at the state-level that occur due to the implementation of new nation-wide welfare programs. The key identifying assumption is that the federal government's decision to introduce a new nationwide scheme is not related to the local economic conditions, including inflation, of specific states. Importantly, the narrative identification does not regard the "broader variations" in state-level transfer spending as exogenous.

In other words, while local macroeconomic conditions may cause systematically higher or lower spending in a state through programs that are well established in the system, it is highly unlikely that a new *nation-wide* program is implemented in response to fluctuations in economic outcomes for a particular state.

Bahal (2019) uses official government records to identify the size, timing, and duration of all spending shocks that occur due to new program introductions between 1980 and 2010 to construct a state-level "narrative shock series". We regard this shock series as a measure of change in government transfers that are largely uncorrelated with the rate of inflation at state-level or any *local* factors that may affect price inflation. Relying on the motivation for program introduction helps rule out reverse causality from inflation to government spending. A residual concern, however, is that regression estimates may still be biased if program spending in a state during program introduction is correlated with key state characteristics that may in turn influence the price dynamics in a state. We check for this and find no correlation between the shock series and key state indices that generally dictate the allocation of program funds from the central government to a state. The results are also robust to the exclusion of state fixed effects which further supports the hypothesis that the shock series is largely independent of state-specific characteristics. These findings are consistent with recent studies on public works in India that find heterogeneity in program implementation to be largely due to idiosyncratic supply-side factors rather than the demand for the program (Imbert and Papp, 2015; Bahal, 2016). Finally, the use of sub-national data allows a causal inference from government transfers to inflation after controlling for national monetary and fiscal policies using year fixed effects.

We hence use a panel data of exogenous shocks in transfer spending for 17 large states over a period of thirty years between 1980 and 2010. Using the

Consumer Price Index for Agricultural Labourers to derive the inflation figures for each state-year, we find no effect of contemporaneous and lagged transfer spending on inflation. Importantly, using the same shock series, Bahal (2019) finds an impact multiplier of 1.4 and a cumulative multiplier of 2.1 for agricultural output at the state-level. This hence suggest that while transfer spending in India due to these welfare programs was expansionary at the local level, it was not inflationary.

As a robustness to our main analysis, we retest our negative results with a different sample and empirical identification. We use the phase-wise implementation of the largest public workfare program in the world: India's National Rural Employment Guarantee Act (NREGA) to estimate a difference-in-differences model to measure whether higher coverage under NREGA was associated with higher price inflation. The program guarantees 100 days of employment per year at minimum wages to every rural household in India. Starting from 2006, NREGA was implemented in a phase-wise manner over three years across all districts of the country. The program covered 200, 330, and all the districts of the country by 2006, 2007, and 2008 respectively. To provide some perspective on the size of the program, during 2009-10, NREGA generated around 2.6 billion work-days and employed around 55 million households. The total expenditure under NREGA amounted to around 0.6 percent of the GDP.

The implementation of NREGA was followed by a period of relatively high inflation in India. This can also be seen in Figure 2 which shows the trend of inflation in India from 2001 onwards. Overheating of the economy due to such large welfare programs is often cited as one of the key reasons for the accelaration in prices during this period. Multiple studies have shown that NREGA did improve economic outcomes, not just for program beneficiaries but also for non-beneficiaries by having a positive general equilibrium effect on real wages in rural India. See Sukhtankar, 2016 for a detailed literature review on the program. However, there has been no direct evidence on how this large public workfare program has impacted price-inflation. We use the phase-wise roll out of the program as a quasi-experiment to check whether the implementation of NREGA led to higher inflation. We find that after controlling for year fixed effects there is no significant association between the implementation of NREGA and inflation. The results again suggest that while government transfers have been successful in raising incomes both at the individual and aggregate levels, they have not resulted in higher prices. The reduced form results suggest two possible underlying mechanisms at play. First, the results may suggest a demand constrained economy that is operating below potential. Alternatively, this suggests government transfers to be potentially expansionary as they boost aggregate demand and alleviate supply constraints as well.

The rest of the paper is organized as follows. In section II, we discuss the narrative analysis following Bahal (2019). This section discusses the construction of the narrative shock series and also conducts various tests to check the validity of the exogeneity assumption. Section 3 and 4 discuss data and empirical model, respectively. Section 5 discusses the results and robustness tests. We conclude in section 6.

2 Narrative Analysis

Following Bahal (2019), we focus on all major rural welfare programs that were implemented by the central government of India between 1980 and 2010.[b] We obtain detailed Information on i) the principal motivation behind the introduction of a new program; ii) the objectives of programs; iii) the implementation

[b]To avoid counting the same expenditure twice, we leave out programs that were subschemes of larger programs.

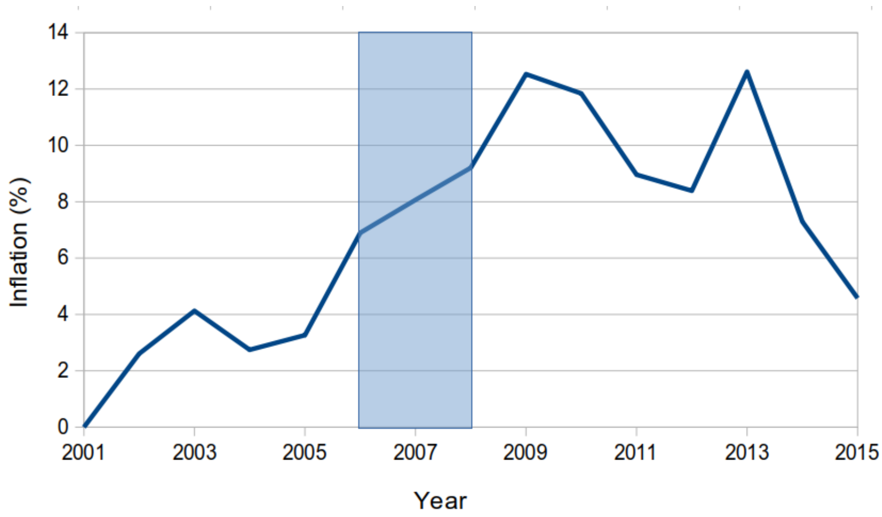

Figure 2: Inflation trend in India
Inflation based on CPI (Agricultural Labourers). The shaded region is the time when NREGA was implemented.

dates of the new programs; and iv) annual state-level expenditure on programs from the *Annual Reports* of the Ministry of Rural Development (MORD).

Generally, most of the expenditure in these programs are in the form of subsidies or wages under public workfares. These are directed toward providing income support to poor families. Such programs can therefore be seen as "conditional transfer" schemes. For example, the work requirement in a workfare program can be seen as targeting mechanism for transfers (Besley and Coate, 1992).

Table 1 provides a summary of all the programs we study. Specifically, the table shows: i) the years during which each program was operational, ii) the type of the program (subsidy or a public workfare), iii) whether the program replaced any older similar program, iv) whether the program was implemented in a phase-wise manner over two years or more, and v) the average annual program expenditure across the whole nation in 2004 prices.

Table 1: Major Rural Welfare Programs since 1980

Program	Years Active	Type of Program	Replaced	Phase-wise Roll-out	Average Annual Exp[†]
IRDP	1978-1998	Credit & Subsidy	None	Yes	₹17.64 Billion
NREP	1980-1988	Public Workfare	None	No	₹19.07 Billion
RLEGP	1983-1988	Public Workfare	None	Yes	₹15.71 Billion
JRY	1989-1998	Public Workfare	NREP & RLEGP	No	₹54.05 Billion
EAS	1993-2000	Public Workfare	None	Yes	₹27.37 Billion
IAY	1995-Present	Housing Subsidy	None	No	₹37.82 Billion
SGSY	1999-Present	Credit & Subsidy	IRDP	No	₹13.99 Billion
JGSY	1999-2000	Public Workfare	JRY	No	₹35.83 Billion
SGRY	2001-2007	Public Workfare	EAS & JRY	No	₹48.91 Billion
NREGA	2006-Present	Public Workfare	SGRY	Yes	₹190.21 Billion

Note: The table shows all major rural welfare programs in India that were operational between 1980 and 2010. IRDP: Integrated Rural Development Program; NREP: National Rural Employment Program; RLEGP: Rural Landless Employment Guarantee Program; JRY: Jawahar Rozgar Yojana; EAS: Employment Assurance Scheme; IAY: Indira Awaas Yojana; SGSY: Swarnjayanti Gram Swarozgar Yojana; JGSY: Jawahar Gram Samridhi Yojana; SGRY: Sampoorna Grameen Rozgar Yojana; NREGA: National Rural Employment Guarantee Act. [†] In 2004 prices. *Source:Bahal (2019)*

Using information on the state-wise annual program expenditure, we define the total rural transfer spending in a state as the total expenditure incurred under all active programs in a given financial year. Figure 3 shows the year-

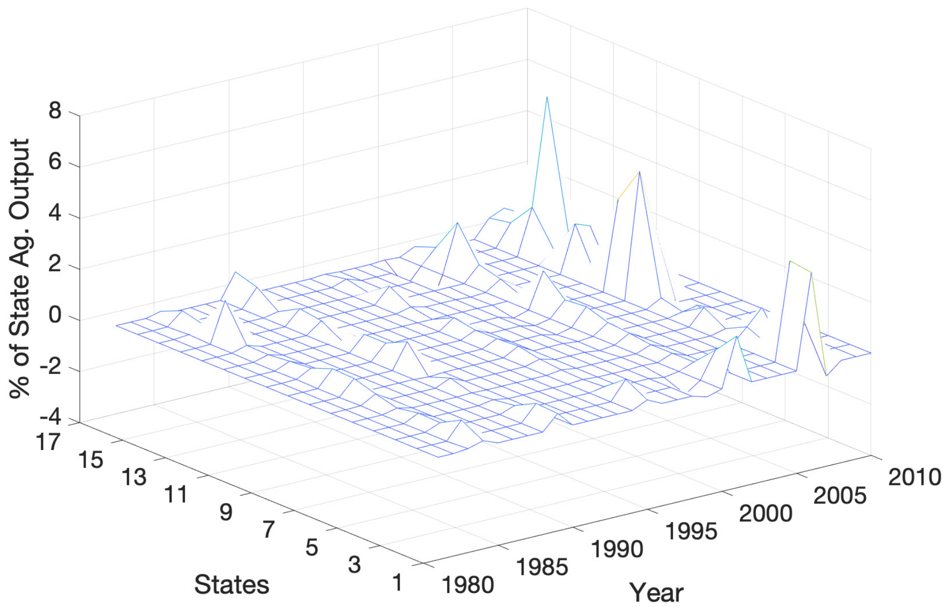

Figure 3: Distribution of expenditure shocks over states and years

Source:Bahal (2019)

on-year change in total rural transfer spending as a percent of previous year's state agricultural output – hereafter referred as the *broader measure* of changes in transfer spending or $B_{i,t}$ – for 17 states between 1981 and 2010. Obviously, this broader measure can include changes in transfer spending that are possibly endogenous to changes in economic activity and, therefore, inflation. In the results section, we compare the impact of transfer spending on inflation obtained using this broader measure of spending changes and the narrative shock series. with those obtained from the broader measure of all changes in transfer spending.

To construct the narrative shock series, we rely on the government documents to identify the motivation behind the introduction of a program. As Bahal (2019) notes "the introduction of a new program is not output or growth motivated" but rather to address inefficiencies in older programs or to address social issues that haven't been addressed by any program before. As a consequence, state-level fluctuations in government transfers that occur due to new program introductions are largely independent of the local economic conditions or inflation dynamics. The next objective is to identify the size, timing, and duration of the spending shocks due to the introduction of a new program.

The timing of a shock is set to the year in which a new program is introduced. See Ilzetzki et al. (2013) and Bahal (2019) for a detailed discussion on why systematic anticipation effects are not likely to be an issue for such government expenditures in developing economies. The duration of a shock is limited to the year in which a new program is introduced. Only for two program introductions: RLEGP in 1983 and NREGA in 2006, the shocks are calculated over two years as these programs were implemented in phases which resulted in a substantial increase in program spending due to the scale-up of the program in its second year

The size of the narrative shocks are the year-on-year change in real per-capita program expenditure that occur due to the introduction of a new program. If a new program replaces older program(s), then the *shock* is simply difference between the new and older program(s). For a standalone program with no predecessor, the whole first-year expenditure is considered as a spending shock. Finally, for the two programs implemented in phases, the year-on-year increase in program expenditure that is associated with the scale-up of the new program is also considered as a spending shock in year t+1.

Narrative Shock Series

Using the narrative sources, every program introduction is thoroughly investigated to check if the introduction of the new program can indeed be deemed independent of local economic conditions. Not all new program introductions meet this criteria and as a result, changes in transfer spending due to the introduction of such programs are not deemed as spending shocks. See Bahal (2019) for a detailed program-by-program discussion on the motivation, timing, and expenditure effects of all new program introductions between 1980 and 2010. Figure 3 shows the state-wise narrative shock series as a percent of state agricultural output for 17 states between 1981 and 2010. As the figure shows, the shock series mostly takes the value zero except for the exogenous impulses that correspond to the introduction of a new program. The size of the spending shocks is consequential as well. On average, the shocks are approximately 0.12 percent of the state agricultural GDP with a standard deviation of 0.51 percent.

We now address some important concerns regarding the exogeneity of the shock series. The primary concern is that stated political motivation behind new programs may in fact not be independent of local economic conditions, although it may be stated otherwise. For example, poor agricultural productivity resulting from low rainfall in past years may lead to a new program for marginal farmers and unemployed agricultural workers. This would imply that the shock series is not exogenous and is predicted by serially correlated shocks to local output. Hence, we check if the constructed shock series is sensitive to past fluctuations in agricultural output and inflation that we regard as proxies for serially correlated shocks. We regress the narrative shock series on contemporaneous and three lags of year-on-year growth in real per capita state agricultural GDP in column one (Table 2). Column 2 reports results after controlling for state and year fixed effects. As can be seen, all coefficients are equal to zero till the second decimal place and are statistically insignificantly different from zero. A test with the null stating that the coefficients of all four regressors are jointly equal to zero yields a p-value of 0.56, hence the null cannot be rejected. Column 3 and 4 report regression results of the shock series without and with state and year fixed effects, respectively. The results are similar to columns 1 and 2, where all estimates are insignificantly different from zero. The joint test of all four coefficients to be jointly equal to zero cannot be rejected with a p-value of 0.53.

Another concern can be that if the increase in spending on new programs is systematically higher for states operating at below potential output then estimates of the effect on inflation may be downward biased. As we show later in the results section, our results are not sensitive to the exclusion of state fixed effects. This confirms that the narrative shock series is largely orthogonal to state-specific characteristics.

Next, we check for the presence of serial correlation in expenditure changes due to the introduction of new programs and find no evidence of it. Conducting the Wooldridge (2002) test for serial correlation in panel data, we cannot reject the null of no serial correlation with a p-value of 0.29. We also regress the shock series on its lag along with state and year fixed effects, and find that the coefficient is insignificantly different from zero. This shows that the expenditure shocks are not predictable using previous shocks.

Finally, we check if political motives were behind program introductions. To check this we use a binary variable $Elec_t$ which takes the value 1 in a general election year and zero otherwise. Again, we do not find any significant correlation of the narrative shock series with $Elec_t$. Hence, the narrative shock series passes a battery of tests which allows us to use expenditure fluctuations due new program introductions as a measure of change in transfer spending that is largely exogenous to price-dynamics at the state-level.

Table 2: Predictability of Narrative Shocks

	(1)	(2)	(3)	(4)
$\pi_{i,t}$	-0.005	0.002		
	[0.005]	[0.008]		
$\pi_{i,t-1}$	-0.004	0.010		
	[0.004]	[0.009]		
$\pi_{i,t-2}$	-0.006	0.008		
	[0.004]	[0.007]		
$\pi_{i,t-3}$	-0.007	0.007		
	[0.006]	[0.005]		
$Y_{i,t}$			0.004	0.002
			[0.002]	[0.002]
$Y_{i,t-1}$			0.003	-0.000
			[0.002]	[0.001]
$Y_{i,t-2}$			0.001	0.000
			[0.001]	[0.001]
$Y_{i,t-3}$			0.004	0.002
			[0.003]	[0.002]
Obs	414	414	459	459

The dependent variable in all regressions is the narrative shock series normalized by lagged state agricultural output $S_{i,t}$. The standard errors reported in square brackets are clustered at the region-year level and are robust to heteroskedasticity.

3 Data

We use government records like the Annual Reports of the Ministry of Rural Development, Government of India to identify all major rural welfare programs that were in operation between 1981 and 2010. These documents contain detailed information for all programs viz. the principal motivation to introduce a new program; the key program objectives; the date of program implementation;

and state-wise financial statements that provide information on annual program expenditure. Agricultural output data at the state level is from the National Accounts Statistics, Central Statistics Office. The state-level inflation variable is constructed using state-specific Consumer Price Index (Agricultural Labour) that are released by the Labour Bureau of India.[c,d] All variables expressed in rupee terms are converted to 2004 prices using the state GDP deflator to control for differential price trends among states.

We use the various decennial censuses of India to get the population numbers. For district-level information on rainfall, we use remote sensed rainfall data from the Tropical Rainfall Measuring Mission (TRMM) satellite. See Fetzer (2020) for a detailed discussion on the consistency and the quality of TRMM data over any other remote sensed or ground-based data.[e] Three new states: Jharkhand, Chhattisgarh, and Uttarakhand were carved from Bihar, Madhya Pradesh, and Uttar Pradesh respectively in 2000. We aggregate all the data of these new states with their respective former states. We hence end up with a panel of 17 large states between 1981 and 2010.

4 Empirical model

We estimate the following empirical model to estimate the relationship between transfer spending and inflation over the period between 1981 and 2010.

$$\pi_{i,t} = \beta_1 S_{i,t} + \beta_2 S_{i,t-1} + \delta_i + \gamma_t + \xi_i t + \varepsilon_{i,t} \tag{1}$$

Here, $\pi_{i,t}$ is the rate of inflation in state i in year t, $S_{i,t}$ is the exogenous expenditure shock as a proportion of state agricultural GDP, measured in real

[c] The CPI numbers were obtained from Indiastat.com.

[d] We also ran regressions using inflation calculated from the GDP deflator and the results were qualitatively the same.

[e] We thank Thiemo Fetzer for sharing the rainfall data.

terms in 2004 prices, δ_i and γ_t are respectively the state and year fixed effects, $\xi_i t$ are state-specific time trends and $\varepsilon_{i,t}$ is the error term. We also run a specification with year-on-year changes in total rural transfer spending as a proportion of agricultural output ($B_{i,t}$) in place of $S_{i,t}$. As noted earlier, $B_{i,t}$ may include changes in transfer spending that are potentially endogenous to local output and prices. Table 3 provides the descriptive statistics for the variables used in equation 1.

Table 3: Summary statistics (equation 1)

Variable	Mean	S.D.	Obs.
$S_{i,t}$	0.13	0.54	510
$B_{i,t}$	0.21	1.00	510
$\pi_{i,t}$	7.38	5.69	465

Note: The number of observations for inflation are lower as the CPI figures for some states are available only from 1995.

In the secondary analysis, where we test the effect of NREGA on inflation, we focus on a more recent sub-sample from 2001 to 2010 where we estimate various versions of equation 2. While the program did not exist before 2006, we take a ten year period around the implementation of NREGA to allow efficient estimation of state-specific heterogenous trends.

$$\pi_{i,t} = \beta_3 \mathbf{I}_{i,t} + \beta_4 \mathbf{I}_{i,t-1} + \mathbf{X}_{i,t} + \delta_i + \gamma_t + \xi_i t + \epsilon_{i,t} \tag{2}$$

In equation 2, $\mathbf{I}_{i,t}$ is a state-level variable denoting the percent of districts in a state where NREGA was implemented. Hence, before 2006 $\mathbf{I}_{i,t}$ takes the value zero for all states. From 2006 onwards, $\mathbf{I}_{i,t}$ increases as the fraction of districts in a state that are covered under NREGA increase. Finally, $\mathbf{I}_{i,t}$ takes the value 100 for all states from 2008 onwards. We construct our specification at the state level as inflation numbers are not available at the district level, and it is also likely that prices within a state are much more correlated than prices across states. We estimate the following specification with all other variables having the same interpretation as equation 1.

$\mathbf{X}_{i,t}$ in equation 2 represents a set of controls that could affect both NREGA implementation and inflation. This includes an indicator for state election year, and the log of average rainfall (in mm) a state receives during rainy season (also referred as monsoon in India). Table 4 shows the summary statistics for the key variables used in equation 2.

Table 4: Summary statistics (equation 2)

Variable	Mean	S.D.	Obs.
$\mathbf{I}_{i,t}$	38.34	44.82	170
$\pi_{i,t}$	6.51	4.41	170
$Rain_{i,t}$	254.35	112.36	170

Note: $Rain_{i,t}$ is the rainfall (in mm) that a state i receives during a rainy season in year t. The statistics for all the variables are calculated across all states and time.

5 Results

The results from estimating equation 1 are given in Table 5. The first column shows the effect of contemporaneous and lagged shocks on inflation, while controlling only for year fixed effects. The coefficient estimates for both contemporaneous and lagged spending shocks is small, negative and insignificantly different from zero. In the second and third columns we add time invariant and time varying state fixed effects, respectively. As the coefficients show, there is no consequential effect of expenditure shocks on inflation in these specifications as well. The coefficients continue to be both statistically and economically insignificantly different from zero. Importantly, comparing the coefficient estimates between the first and third column suggests that the impact of spending shocks on inflation is fairly independent of time invariant and time varying state fixed effects. This reaffirms our premise that the spending shocks are orthogonal to state characteristics. Finally, in the column 4, the key explanatory variables are contemporaneous and lagged values of all changes in transfer spending in a state ($B_{i,t}$). The coefficients continue to be insignificant in this specification.

The null result of the narrative shock series on inflation is remarkable especially because Bahal (2019), using a similar specification, shows spending shocks to increase state agricultural output quite substantially. Bahal (2019) reports an impact multiplier of 1.4 which increases to 2.1 over a period of five years. Put together, the results suggest that while changes in transfer spending due to new program introductions are expansionary, they are not necessarily inflationary. This is indicative of an economy operating well below potential output where redistributive fiscal transfers increase aggregate demand and output without building inflationary pressures.

For robustness, we confirm our results using the difference-in-differences setup using the phase-wise implementation of NREGA over the period from 2001 to 2010. Table 6 shows the results of estimating equation 2. All columns show the coefficients of the contemporaneous and lagged NREGA implementation variable. While the first column estimates equation 2 without any additional controls, the second column controls for state fixed effects. The contemporaneous coefficients are statistically and economically significant in both these columns showing a strong correlation between the implementation of NREGA and inflation. In both columns 1 and 2, the coefficient on $I_{i,t}$ suggests that a state fully covered under NREGA would have had 7 percentage point higher price inflation relative to a state without NREGA. Prima facie, these results appear contradictory to our findings over the larger sample using the narrative shock series.

Table 5: Inflation and Transfer Spending 1980-2010

	(1)	(2)	(3)	(4)
$S_{i,t}$	-0.212	-0.155	-0.040	
	[0.224]	[0.250]	[0.255]	
$S_{i,t-1}$	-0.250	-0.194	-0.038	
	[0.170]	[0.196]	[0.198]	
$B_{i,t}$				-0.054
				[0.164]
$B_{i,t-1}$				0.068
				[0.153]
Obs.	451	451	451	451
Year FE	Yes	Yes	Yes	Yes
State FE	No	Yes	Yes	Yes
Trends	No	No	Yes	Yes
Obs.	451	451	451	451

The dependent variable in all regressions is $\pi_{i,t}$: the year-on-year rate of inflation for a state i in year t. The standard errors reported in square brackets are clustered at the region-year level and are robust to heteroskedasticity.

However, adding year fixed effects in the third column render the coefficients statistically insignificant. Adding the state-specific time trends together with other controls in column 4 do not change the results obtained in column 3. Even after disregarding the standard errors and statistical insignificance, the contemporaneous and lagged estimates of **I** in column 4 add to 0.005. This implies that, relative to the eight percentage point increase in inflation between 2005 and 2010 (see figure 2), full implementation of NREGA in a state increased inflation by an economically insignificant amount of 0.5 percentage points over two years. The p-value for testing if the sum of the two coefficients in column 4 is zero is 0.62 with an F-statistic of 0.25. The remarkable change in the size and significance of the coefficient of $\mathbf{I}_{i,t}$ underscores the influence of aggregate shocks in determining price dynamics during this period. Hence, the strong correlation between inflation and program implementation in the first two columns in Table 6 can be attributed to aggregate country-level factors such as a rise in international crude oil prices (see Mohanty and John, 2015) and not to the implementation of the program itself. In conclusion, the absence of any association between transfer spending and inflation obtained using narrative analysis continues to hold with a different empirical model and sample.

Table 6: NREGA and Inflation

	(1)	(2)	(3)	(4)
$\mathbf{I}_{i,t}$	0.070***	0.070***	0.024	0.026
	[0.007]	[0.007]	[0.013]	[0.014]
$\mathbf{I}_{i,t-1}$	0.002	0.003	-0.018	-0.021
	[0.008]	[0.008]	[0.019]	[0.018]
State F.E.	No	Yes	Yes	Yes
Year F.E.	No	No	Yes	Yes
Trends[a]	No	No	No	Yes
Obs.	153	153	153	153

The dependent variable in all the regressions is $\pi_{i,t}$: the year-on-year rate of inflation for a state i in year t. The standard errors reported in square brackets are clustered at the region-year level and are robust to heteroskedasticity. $^{*}p < 5\%$, $^{**}p < 1\%$, $^{***}p < 0.1\%$

[a] : also includes other controls like an indicator for state election year and (log) rainfall at the state-level.

6 Conclusion

Using a narrative shock series constructed from data on government transfers via welfare programs in India over a period of 30 years, we show that fiscal transfers have not been inflationary. Importantly, the same fiscal transfers have been shown to increase output. As a robustness check, we focus on a recent large public workfare in India that has been shown to increase real wages and is suspected of causing inflation. Exploiting the phase-wise implementation design of the program, we show that the implementation of this program is not associated with higher inflation. Hence, we conclude that welfare programs which consist primarily of fiscal transfers, have not caused inflation in India.

These results have obvious policy implications. Well targeted transfers through government safety-net programs can be a potent redistributive fiscal tool that can be used to stimulate the economy without the trade-off of higher inflation. However, it is important to note that the results noted in this study are reduced form estimates. There are two possible mechanisms that can explain the absence of a positive relationship between transfer spending and inflation in India. First, it is possible that output at the state level has always been below potential output. The second channel suggests that transfer spending has not only boosted aggregate demand but have also had positive supply-side affects. There is suggestive evidence in the literature to support this (Bahal, 2019; Basu, 2013; Muralidharan et al., 2016a,b). Future work can be focused on understanding whether such programs have indeed had productivity enhancing effects at the local level.

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